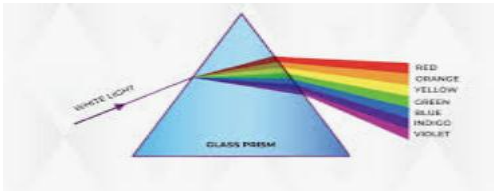
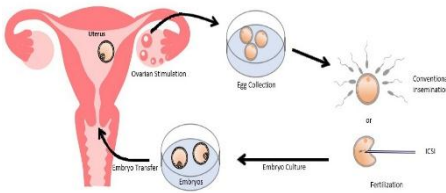
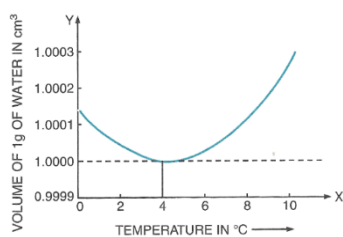


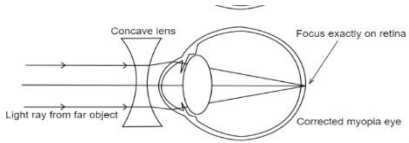
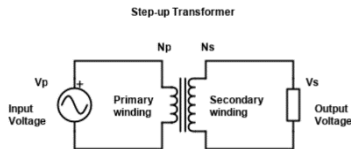
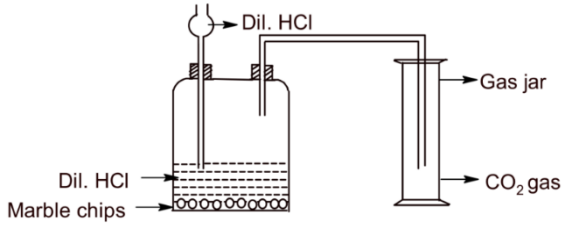
SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुञ्जिका

यस उत्तर कुञ्जिका र मार्किङ्गस्किममा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ । उत्तरकुञ्जिकामा प्रत्येक चरणको प्राप्ताङ्क १ प्रदान गर्नुपर्नेछ तथा एकाइ छुट भएमा अङ्क नघटाउने ।

प्र. नं.	उत्तर	अङ्क
खण्ड “क” Group A		
1) a.	iii. Excessive hunger and thrust	1
b.	iv. Malabar nut, calamus, Caterpillar fungus	1
c.	i. Burning of fossil fuels	1
d.	ii. 490 N	1
e.	ii. Pascal's law	1
f.	iii. Magnetic to electrical energy	1
g.	ii. $1s^2, 2s^2, 2p^6, 3s^2$	1
h.	i. Be, Mg, Ca	1
i.	i. in beaker A and B exothermic and C endothermic reaction has occurred	1
j.	ii. Siderite	1
खण्ड “ख” Group B		
2) a.	Amount of solvent	1
b.	 or similar	1
c.	Number of RBC is less than normal	1
d.	Any one of the following - Poaching - Lack of proper conservation - Habitat loss - or any other appropriate answers	1
e.	11 N	1
f.	Total internal reflection	1
g.	They react with water form alkaline solution/metallic hydroxide	1
h.	Metallic property decreases from left to right	1
i.	Heating impure metal in presence of excess oxygen to make pure metal	1
खण्ड “ग” Group C		
3.	$(ms^{-1})^2 = (ms^{-1})^2 + (ms^{-2})m^2$ or, $m^2s^{-2} = m^2s^{-2} + m^3s^{-2}$ Since the units are not homogeneous, the equation is incorrect.	1 1
4.	One mark each for any two elements - Obtain information - Learn problem solving techniques - Watch videos, simulations, pictures - or any other suitable answer	1 1

5.	- Select the point where the video is to be cut (cropped from) and split - Delete the unwanted clip and save the video Or any other appropriate answer		1 1
6	One mark each for two points - It does not have spines - body not radially symmetrical - or any other suitable answer		1 + 1
7	One mark each for two points - Presence of vascular bundle - Root, stem and leaves are prominent - or any other suitable answer		1 1
8.	One mark each for two points - economic - pollination - medicinal Or any other suitable answer		1 1
9.			1 + 1
10.	DNA	RNA	1 1
	Adenine, Guanine, Thymine, Cytosine	Uracil at the place of Thymine	
	Transfer of genetic material	Protein synthesis	
Or any other correct answer			
11.	$F = G \frac{m_1 m_2}{d^2}$ $F = 2 \times 10^{11} \text{ N}$ No mark reduction for not writing the unit		1 1
12.	Free fall	Weightlessness	1+1
	Falling of objects in absence of resistance	Weight of object at free fall Or the feeling of not having Weight	

13.			2									
14 a	The universe will end in a big crunch		1									
14 b	Line Y		1									
15.	$2Al(s) + 3H_2SO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3H_2(g).$		1 1									
16.	Soap	Detergent	1									
	Sodium or potassium salt of fatty acid	Synthetic petrochemical obtained from hydrocarbon	1									
	Biodegradable	Not biodegradable										
	Or any appropriate answer											
खण्ड “घ” Group D												
17	(a) IVF is a technique in which an egg and sperm are combined outside the body (in a lab) to form a zygote, which is then implanted into the uterus to achieve pregnancy.  (b)- <table><tr><th>Feature</th><th>Inbreeding</th><th>Line Breeding</th></tr><tr><td>Degree of Relation</td><td>Very close relatives (e.g., parent-offspring, brother-sister)</td><td>Distant relatives, usually to preserve traits from a superior ancestor</td></tr><tr><td>Genetic Risk / Purpose</td><td>High risk of genetic defects; mainly to fix traits quickly</td><td>Lower risk; to maintain desirable traits safely</td></tr></table>		Feature	Inbreeding	Line Breeding	Degree of Relation	Very close relatives (e.g., parent-offspring, brother-sister)	Distant relatives, usually to preserve traits from a superior ancestor	Genetic Risk / Purpose	High risk of genetic defects; mainly to fix traits quickly	Lower risk; to maintain desirable traits safely	1+1 =2 1+1= 2
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18	(a) Leukopenia (b) Platelets (c) Anemia (d) B		1 1 1 1									
19	(a) Water- density first increases upto 4°C and then decrease but in case of other substances density decreases on increasing temperature from 0°C (b) allowing aquatic life to survive in winter. (c) Graph  (d) Water cannot be used as a thermometric liquid because of its anomalous expansion.		1 1 1 1									

20	(a) Myopia (near /shortsightedness) (b) The eyeball is too long or the cornea is too curved, so light focuses in front of the retina instead of on it. (c) corrected diagram  (d) Convex lens	1 1 1 1
21	(a) use the $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ relation and calculate secondary voltage $V_s = 2200\text{ V}$ (b) Step up transformer 	1 1 1 1
22	(a) fig  Fig: Laboratory preparation of CO₂ gas (b) reaction On passing CO₂ for a long time – formation of soluble calcium bicarbonate: $\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{HCO}_3)_2$ (c) In photosynthesis	1 + 1 1 1
23	(a) Homologous series – A group of organic compounds that have the same functional group, similar chemical properties and each successive member differs by –CH₂ – unit (b) (i)Propane (C₃H₈) (ii) Ethyl alcohol (Ethanol) $\begin{array}{c} \text{H} & & \text{H} \\ & & \\ \text{H} - \text{C} - & \text{C} - & \text{OH} \\ & & \\ \text{H} & & \text{H} \end{array}$ (iii) C_nH_{2n}	1 1 1 1